TANZANIA HEADS OF ISLAMIC SCHOOLS COUNCIL

FORM FOUR INTER ISLAMIC MOCK EXAMINATION

CHEMISTRY 1

032/1

(For both School and Private Candidates)

TIME: 3 Hours **Wednesday, 23rd August 2023 a.m.**

Instructions

1. This paper consists of three sections A, B and C with a total of **eleven (11)** questions.
2. Answer all questions in sections A and B and only two (2) questions for section C.
3. Section A carries sixteen (16) marks, section B carries fifty-four (54) marks and section C carries thirty (30) marks.
4. Non-programmable calculators may be used.
5. Cellular phones and any unauthorized materials are not allowed in the Examination Room.
6. Write your Examination Number on every page of your answer booklet(s).
7. The following constants may be used:

   Atomic masses: H=1, C=12, N=14, O =16, Na=23, Cl=35.5, Ca=40, Mg=24, S=32, Ag =108, Cu =64, Fe =56, Pb =208

   Avogadro's number (L_A) = 6.02×10^{23}

   GMV at s.t.p. = 22.4 dm^3/mol

   1 Faraday = 96500 Coulombs

   1 litre = $1dm^3 = 1000cm^3$.

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SECTION A (16 Marks)

Answer **all** questions in this section

1. For each of the item(s) (i) – (x), choose the correct answer from among the given alternatives and write its letter beside the item number in the answer sheet(s) provided.

   (i) Calcium chloride crystals on a watch glass left overnight in the laboratory. Eventually a solution was found in the watch glass, this suggests that this compound is:
   A. Efflorescent
   B. Hygroscopic
   C. Deliquescent
   D. Solute
   E. Sublime

   (ii) Over seven (7) billion people on the Earth use water everyday yet Earth's water supply remains relatively constant. Which statement best explains this?
   A. The sea level is rising
   B. Water is constantly recycled by hydrological cycle
   C. Global warming melts ice to replace water that is used
   D. Water exists in three phases on Earth
   E. Water is a universal solvent

   (iii) A form one student went in the chemistry laboratory, accidentally he spilled concentrated sulphuric acid (H_2SO_4) on his skin, which action do you think should be taken immediately by this student?
   A. Rinse off with large quantity of running water
   B. Neutralize with solid $CaCO_3$
   C. Neutralize with concentrated NaOH
   D. Neutralize with concentrated HNO_3
   E. Rinsed with dilute sulphuric acid

   (iv) An element in the periodic table with atomic number 18 belongs to which of the following?
   A. Group I and period 1
   B. Group II and period 3
   C. Group VIII and period 3
   D. Group V and period 4
   E. Group VII and period 4

   (v) When chlorine gas is bubbled through distilled water, the distilled water turn pale yellow. Which of the following equations represents the reaction?
   A. $Cl_{2(g)} + H_2O_{(l)} \longrightarrow Cl_2.H_2O_{(aq)}$
   B. $Cl_{2(g)} + H_2O_{(l)} \longrightarrow HCl_{(aq)} + HOCl_{(aq)}$
   C. $2Cl_{2(g)} + 2H_2O_{(l)} \rightleftharpoons 4Cl_{(aq)} + 2H_2O_{(aq)}$
   D. $Cl_{2(g)} + 2H_2O \rightleftharpoons 2HOCl_{(aq)} + H_{2(g)}$
   E. $Cl_{2(g)} + 2H_2O \longrightarrow 2HCl + O_2$

(vi) When the teacher finished his session, he left a piece of chalk on Nassor's table. Nassor started to grind that piece of chalk until it became powder. The following statements are correct about the changes observed except:

A. The piece of chalk can be recovered
B. No new substance is formed
C. Energy is produced or absorbed when grinding the chalk
D. There is no change in mass
E. The components of piece of chalk are not affected

(vii) Mr. Chorogondo went to the fuel factory to buy a fuel that will enable his wife to use at home. Being an expert, you will advise Mr. Chorogondo to choose a fuel with:

A. Low pyrometric burning effect
B. Low ignition point
C. High velocity of combustion
D. High content of non-combustible materials
E. Low content of non-combustible materials

(viii) During a practical session, a student reacted aqueous solution of silver nitrate with aqueous solution of sodium chloride. Insoluble silver chloride and aqueous solution of sodium nitrate were formed. This reaction is said to be:

A. Decomposition reaction
B. Displacement reaction
C. Precipitation reaction
D. Redox reaction
E. Combination reaction

(ix) Kilango and Kimbi were arguing about the general formula of an alkyl group. Which among the following do you recommend to them?

A. CnH_2n
B. CnH_2n-1
C. CnH_2n+2
D. CnH_2n+1
E. CnH_2n-2

(x) At Taifa road junction, a truck carrying kerosene gets an accident, and unfortunately fire breaks out. People wanted to extinguish it using water but they were told not use water because water would spread the fire. This is because:

A. Water is denser than kerosene
B. Water is less dense than kerosene
C. Water floats over the kerosene
D. Water is neutral to fire
E. Water is a universal solvent

2. Match the effects of various chemicals on the living body in list A with their corresponding warning signs in list B by writing the letter of the correct response beside the item number in the answer booklet provided.

LIST A		LIST B
(i)	When poured on the wool, it completely damages the surface of the wool	A. Toxic B. Radioactive materials C. Corrosive D. Flammable E. Oxidant F. Explosive G. Irritant H. Fragile
(ii)	When poured on the small fire, it bursts and turns a bigger fire	
(iii)	When brought near the skin of someone, he/she starts to complain that his/her skin is itching	
(iv)	When exposed on air, sudden blast accompanied with light occurs.	
(v)	When poured into the food, the rat that eats it dies	
(vi)	When exposed to you for a longtime, your organs will be damaged	

SECTION B (54 Marks)

Answer all questions in this section

3. (a) We encounter many events/processes in our daily life that are good examples of chemical changes. Give three (3) examples of these changes.

   (b) List down three properties that clearly justify whether muddy water is a mixture or a compound.

   (iii) Iron (III) chloride is an electrovalent compound which ionizes in water when electric current is allowed to flow through it.

   (i) How many moles of ions are formed when the compound is completely ionized in water?

   (ii) How many ions will be responsible for the passage of electric current in it?

4. (a) Form four students were taken to the chemistry laboratory for demonstrations of organic reactions between organic acids and alcohol. They found on the bench two reagents; butanoic acid and ethanol.

   (i) Write a chemical equation for this reaction

   (ii) Give the IUPAC name of the major product formed

   (iii) Which name is assigned to this type of reaction?

   (b) A hydrocarbon contains 80% carbon and 20% hydrogen by mass. Its' vapour density was measured and found to be 15. Calculate:

   (i) Its empirical formula

   (ii) Its molecular formula

   (iii) Which family of hydrocarbon does the molecular formula belong?

5. Lack of safe and clean water for industrial and domestic purposes is a serious problem in most urban and rural areas in Tanzania. One of the major causes of this is the pollution of sources of water due to the addition of soluble compounds of calcium and magnesium

   (a) Explain three methods that could make such water useful

   (b) (i) Give three economic disadvantages of using such water domestically and industrially.

   (ii) State at least one advantage of using such water.

   (c) Food substances are broken down by enzymes in the stomach before being absorbed by blood. Sometimes this process of breaking it down is slowed down resulting into in-digestion and stomach up-set. From the chemistry point of view, state out two factors that could be the cause of such problem.

6. (a) A universal indicator has a pH values ranging between 1-14. Five solutions P, Q, R, S and T were subjected to it and their pH values read as follows:

Solution	P	Q	R	S	T
pH	01	05	07	09	14

   Which solution was:

   (i) Strong base

   (ii) Strong acid

   (iii) Weak acid

   (iv) Neutral solution

   (v) Weak base

   (b) A metal X (atomic number 11) burns in chlorine to produce a white solid chloride Y

   (i) By means of diagrams illustrate the arrangement of electrons in X before and after the reaction.

   (ii) Write a chemical equation for the reaction

7. (a) A current of 0.5A is allowed to flow for six minutes and twenty seconds through two cells connected in series. If at the end of the reaction 0.216g of silver and 0.659g of nickel were deposited at their respective electrodes. Calculate the:

   (i) Quantity of electricity passed through the two cells

   (ii) Quantity of electricity that would be needed to electroplate a metal bar with 1g of silver.

   (iii) Quantity of electricity that will be needed to deposit 1g of nickel atoms.

   (b) You are provided with the symbols of elements with atomic number 6-17 inclusive

   C, N, O, F, Ne, Na, Mg, Al, Si, P, S, Cl

   (i) Which two atoms of these elements will form ions of the same electronic configuration?

   (ii) Give the name and formula of one compound that will contain three of these elements.

   (iii) Choose two elements that will form a soluble acidic oxide and write its formula and the name.

8. (a) With the help of the diagram briefly describe the three stages involved in the extraction of Sulphur from its Ore.

   (b) State briefly the essence of using hot compressed air and super-heated steam of $170^{o}C$ in the process in 8(a) above.

SECTION C (30 Marks)

Answer only **two (2)** questions in this section

9. A soil may be fertile but not productive. Discuss (five points)

10. Extraction of metal is a reduction process. Explain briefly using two examples of metals above carbon and below carbon in the series about this fact.

11. Write an essay about environmental conservation using the following guidelines:

    (i) Meaning of environmental conservation

    (ii) Importance of environmental conservation (three points)

    (iii) Personal involvement in conserving the environment (four points)

Wabillah Tawfiq